

SAMARTH PRATAP SINGH

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🔗 [LeetCode](#)

🗣️ [Hugging Face](#)

EDUCATION

VIT Bhopal University, Bhopal

2023 – 2027

B.Tech - Computer Science and Engineering - **CGPA - 8.45**

Bhopal, Madhya Pradesh

COURSEWORK

- DSA
- OOP Concepts
- Cloud Computing
- DBMS
- Operating Systems
- Computer Networks
- Software Engineering

PROJECTS

Argus - Autonomous Research Engine 🚀 | [FastAPI](#), [LangGraph](#), [Grog](#), [Docker](#), [LangSmith](#) Feb 2026

- Built a production-grade multi-agent research pipeline using LangGraph supervisor pattern - 5 specialist agents (planner, researcher, critic, writer, supervisor) orchestrated via LLM-driven `Command(goto)` routing.
- Designed async job architecture (submit → poll → fetch) with SQLite persistence and LangGraph checkpointing; research jobs survive agent failures and every LLM call is traced end-to-end in LangSmith.
- Containerized with Docker, deployed on Render with a health-check endpoint; integrated Tavily web search, ArXiv, and Wikipedia as agent tools to synthesize cited markdown reports in 30–90 seconds.

DoCopilot (RAG Document Q&A) 🚀 | [Next.js](#), [FastAPI](#), [Qdrant \(Hybrid\)](#), [Reranking](#), [Grog](#) Dec 2025

- Built a full-stack RAG app to upload PDFs/TXT, index content using Qdrant hybrid search (BM25 + dense vectors), and answer questions with citations.
- Improved retrieval quality via RRF fusion + cross-encoder reranking; selected top-k contexts before generation to reduce noise.
- Added guardrails (prompt-injection detection, PII redaction, source-grounding checks) and an evaluation + ablation study with LLM-as-Judge; reported 89.2% avg correctness, 90.5% relevance, 100% source rate, and 2.86s avg latency on 40/40 queries.

FLAN-T5 Dialogue Summarizer (GenAI) 🚀 | [PEFT](#), [LoRA](#), [Transformers](#), [Gradio](#) Oct 2025

- Fine-tuned FLAN-T5-base with LoRA on SAMSum (14.7K dialogues), achieving 49.01 ROUGE-1, 72.25 BERTScore F1, and 42.51 METEOR.
- Implemented parameter-efficient training (LoRA $r=16$, $\alpha=32$), updating only 2% of parameters with FP16 mixed precision for faster, memory-efficient training.
- Deployed an interactive Gradio app on Hugging Face Spaces with configurable decoding (beam search, max length) and published the model with reproducible evaluation (ROUGE, BERTScore, METEOR, BLEU).

RoBERTa for Banking Intent Classification(Banking77) 🚀 | [PyTorch](#), [Transformers](#), [CUDA](#) Sep 2025

- Fine-tuned RoBERTa-base on Banking77 (77 intents, 13k queries) to 93.7% accuracy and 93.6% macro-F1 on a held-out split, with per-epoch tracking and best-checkpoint selection for robust evaluation.
- Implemented standard transformer fine-tuning with AdamW and weight decay ($2e-5$ LR; batch 16/32; 5 epochs), using FP16 on a GPU-accelerated Linux environment for efficient training.
- Added experiment hygiene: fixed seeds, consistent tokenization/padding, and epoch-level metrics; verified inference with a minimal loading script to streamline reviewer replication.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, SQL

ML/DL: PyTorch, TensorFlow, Transformers, PEFT/LoRA

LLM/Agent Systems: LangGraph, LangChain, RAG, Prompting, Tool-Calling

Retrieval: Qdrant, FAISS, Hybrid Search (BM25 + Dense), Reranking

MLOps/Observability: LangSmith, MLflow, Weights & Biases

Backend/Prod: FastAPI, Docker, Linux, Git/GitHub

Full-Stack: Next.js, PostgreSQL, NextAuth.js

CERTIFICATIONS

- Applied Machine Learning in Python - Coursera